

# Production Order Create/Edit — Basic Design Document (基本設計書)

## Document control (改版履歴)

| Field                      | Value                  |
|----------------------------|------------------------|
| Document ID                | 001_BD                 |
| Category                   | UI                     |
| System name                | ProductionManagementAI |
| Subsystem name             | Production orders      |
| Work item                  | WI-002                 |
| Based on brief.md revision | 1                      |
| Created by                 | Claude (for ThanhTN)   |
| Created date               | 2026-09-18             |
| Last updated by            | Claude (for ThanhTN)   |
| Last updated date          | 2026-09-23             |

| Version | Date       | Author               | Revision content                                                                                                                                          |
|---------|------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | 2026-09-18 | Claude (for ThanhTN) | Initial creation (WI-002 restart against the rewritten BD template; supersedes the scrapped first-pass 001_BD)                                            |
| 2       | 2026-09-18 | Claude (for ThanhTN) | DEC-009–DEC-012 resolved: due-date check condition, stale-save rejection (V-08), plant timezone, PO-YYYY-NNNNN order number                               |
| 3       | 2026-09-18 | Claude (for ThanhTN) | DEC-017–DEC-020: plant timezone Asia/Tokyo ; Cancel confirmation when edited (REQ-019, items 17–19, E-07/E-09); 30 seeded products; CSRF approach decided |
| 4       | 2026-09-18 | Claude (for ThanhTN) | Aligned with 001_DD: quantity upper bound (V-02, DEC-024)                                                                                                 |
| 5       | 2026-09-22 | Claude (for ThanhTN) | Screen B exists (WI-003): the screen's entry point and every exit that used to lead to the home page now lead to the production-order list                |

| Version | Date       | Author               | Revision content                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------|------------|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6       | 2026-09-22 | Claude (for ThanhTN) | Completion tracking for Screen C (WI-004 REQ-033, DEC-002/DEC-003): saving <code>InProgress</code> → <code>Completed</code> records the completion time (FN-006, Actions and business rules, Data design overview). No visible change to SCR-001                                                                                                                                                                                                                    |
| 7       | 2026-09-22 | Claude (for ThanhTN) | WI-004 (DEC-016, DEC-021): the shared header now carries the application navbar specified in 003_BD "Shared application header"; items 1–2 point to it. The breadcrumb stays. WI-004 DEC-022: the discard-changes confirmation (FN-009, dialog 17) now also guards leaving an edited form through any in-app link — navbar, breadcrumb, app name — not only Cancel (E-07a); Discard then goes to that link's destination. No field, rule, validation or API changes |
| 8       | 2026-09-23 | Claude (for ThanhTN) | Diagrams redrawn for the revised BD template (RFC 0009): screen transition and status state machine in Mermaid; PC and SP layouts as SVG wireframes under <code>wireframes/</code> . The screen transition now matches the screen list: Cancel returns to SCR-002, which the text diagram had not shown since version 5. No field, rule, validation, event or API changes                                                                                           |
| 9       | 2026-09-23 | Claude (for ThanhTN) | WI-005: the UI is Japanese. Quoted UI text and messages give the implemented Japanese text with an English gloss (WI-005 DEC-003); status labels (M-01, M-02), timestamp format (M-04, <code>YYYY/MM/DD HH:mm</code> ), page titles and wireframes updated; demo products are automobile parts with Japanese names; two screen-transition edges drawn longer so their labels don't overlap in the PDF. No field, rule, validation, event or API changes             |

## System overview

One screen, SCR-001, lets a signed-in Admin or Operator create a new production order or edit an existing one, including moving it through its status workflow. It is the harness's first end-to-end business screen ("Screen A").

| Requirement ID | Description                                 | Covered by section                                                                 |
|----------------|---------------------------------------------|------------------------------------------------------------------------------------|
| REQ-010        | Create a production order                   | Business flow; §3 screen items; §6 events; Success and exception flows             |
| REQ-011        | Edit an existing production order           | Business flow; 0-3 URL parameters; §5 V-08; §6 events; Success and exception flows |
| REQ-012        | Only authenticated Admin/Operator           | 0-1 basic information; Actions and business rules; Exception flows                 |
| REQ-013        | Quantity positive whole number              | §5 validation V-02                                                                 |
| REQ-014        | Due date today or later                     | §5 validation V-03, V-04                                                           |
| REQ-015        | Product must exist                          | §5 validation V-01; Data design overview                                           |
| REQ-016        | Notes ≤ 500 characters                      | §5 validation V-05                                                                 |
| REQ-017        | Fixed status state machine                  | Screen transition (status diagram); §3 item 6; §4 M-02; §5 V-06                    |
| REQ-018        | Product/quantity locked after Draft         | §3 display conditions; §5 V-07; Exception flows                                    |
| REQ-019        | Confirm before discarding changes on Cancel | §3 items 17–19; §6 E-07, E-09                                                      |

# Overall configuration and architecture

Uses the confirmed stack ( ai/project.md ): a React (Vite + TypeScript + Tailwind CSS v4) page calling the .NET 10 backend's JSON API over the same origin, layered Domain/Application/Infrastructure/Api (0001\_ADR), persisted in PostgreSQL 17. Access is gated by WI-001's cookie session and role checks (0002\_ADR): ProtectedRoute on the frontend, an authorization policy on the backend. No new architecture boundary or external system is introduced.

## Function list

| Function ID | Function name                   | Description                                                                                                        | Related requirement ID             |
|-------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------|------------------------------------|
| FN-001      | Create production order         | Save a new order with a system-generated order number and status Draft                                             | REQ-010                            |
| FN-002      | Edit production order           | Save changes to an existing order's editable fields                                                                | REQ-011, REQ-018                   |
| FN-003      | Load production order           | Fetch an existing order for display in edit mode                                                                   | REQ-011                            |
| FN-004      | List products                   | Provide the product choices for the product dropdown                                                               | REQ-015                            |
| FN-005      | Validate order input            | Enforce product, quantity, due date and notes rules (client for feedback, server authoritative)                    | REQ-013, REQ-014, REQ-015, REQ-016 |
| FN-006      | Enforce status transitions      | Allow only the defined state-machine edges; on InProgress → Completed , record the completion time (WI-004 FN-021) | REQ-017; WI-004 REQ-033            |
| FN-007      | Enforce field lock by status    | Reject product/quantity changes once the order is not Draft                                                        | REQ-018                            |
| FN-008      | Enforce authentication and role | Reject unauthenticated (401) and non-Admin/Operator (403) requests                                                 | REQ-012                            |
| FN-009      | Confirm discard on leaving      | Ask before discarding unsaved changes, whether the user leaves by Cancel or by an in-app link (WI-004 DEC-022)     | REQ-019                            |

## Actors and business flow

Actors: Admin, Operator — both already signed in through WI-001's login screen (DEC-001).

### Create (UC-001)

1. User opens the create route → system loads the product list (FN-004) and shows an empty form; status shows `Draft` , order number shows 「保存時に採番」 (Assigned on save).
2. User selects a product and enters quantity, due date and optional notes → system shows field errors as fields lose focus (FN-005).
3. User presses **保存** (Save) → system validates all fields client-side; if valid, sends the request; the server re-validates (FN-005, FN-008) and saves (FN-001).
4. On success → system navigates to the edit route of the new order and shows 「製造指示 {order number} を登録しました。」 (Production order {order number} created.)

### Edit (UC-002, UC-003)

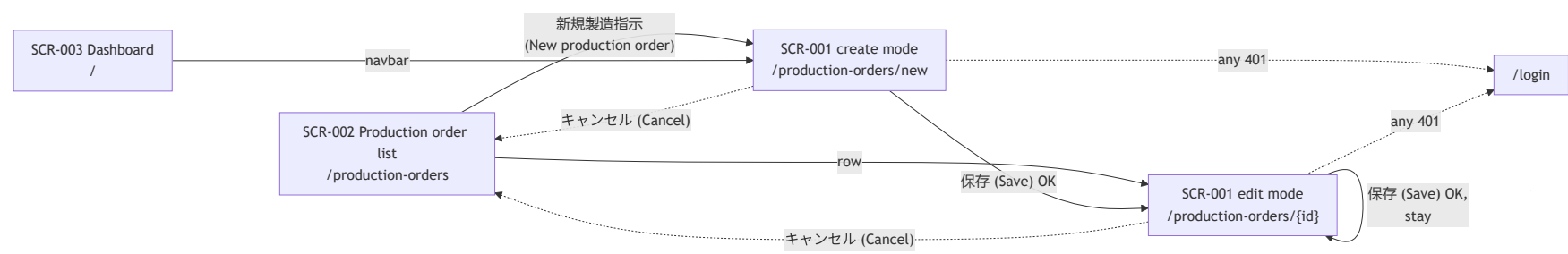
1. User opens the edit route for an order → system loads the order (FN-003) and products (FN-004) and shows the form pre-filled; product/quantity are read-only unless status is `Draft` ; the status dropdown offers only the current status and its valid next states (DEC-008).
2. User changes editable fields and/or status → client-side validation as above.
3. User presses **保存** (Save) → server re-validates (FN-005–FN-008) and saves (FN-002).
4. On success → form shows the saved values and 「製造指示 {order number} を保存しました。」 (Production order {order number} saved.)

**Cancel (REQ-019): キャンセル** (Cancel) returns to the production-order list, SCR-002 ( `/production-orders` ), without saving. If the form has unsaved changes, the system first asks 「変更を破棄しますか？」 (Discard your changes?) (FN-009); **編集を続ける** (Keep editing) returns to the form with all values kept.

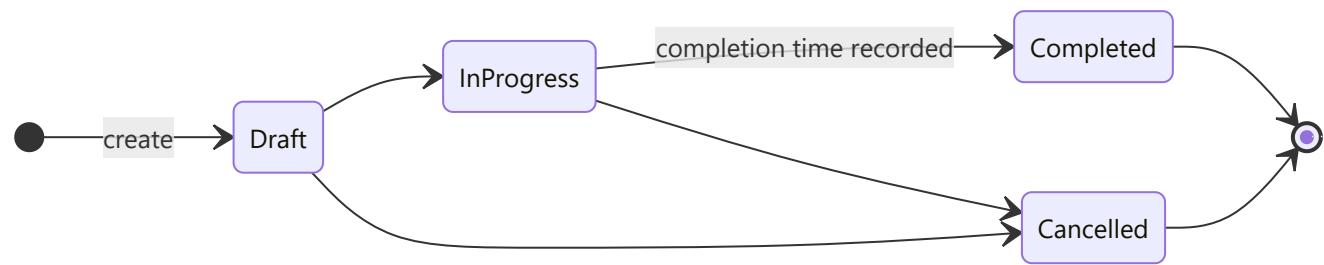
## Screen list and screen transition

| Screen ID | Screen name                  | Entry point                                                                                                                                                                                                                       | Exit / next screen                                                                                                                                                                    |
|-----------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCR-001   | Production Order Create/Edit | Create mode: <code>/production-orders/new</code> , from the 「新規製造指示」 (New production order) action on SCR-002 or on the home page. Edit mode: <code>/production-orders/{id}</code> , from a row on SCR-002 (or the direct route). | Save (create) → SCR-001 edit mode for the new order. Save (edit) → stays on SCR-001. Cancel → SCR-002 <code>/production-orders</code> . Session expired / 401 → <code>/login</code> . |

Screen transition:



Order status state machine (REQ-017, DEC-003):



Completed and Cancelled are terminal.

## Screen design detail

### SCR-001 Production Order Create/Edit

#### 0-1. Basic information (基本情報)

| No | Item          | Content                                                         | Reference                |
|----|---------------|-----------------------------------------------------------------|--------------------------|
| 1  | Route / path  | /production-orders/new (create), /production-orders/{id} (edit) | 0-3                      |
| 2  | API base path | /api/production-orders , /api/products                          | Exact contract in 001_DD |

| No | Item                             | Content                                                                                            | Reference       |
|----|----------------------------------|----------------------------------------------------------------------------------------------------|-----------------|
| 3  | Character encoding               | UTF-8                                                                                              |                 |
| 4  | Error page / fallback            | Load failure or unknown {id} : in-page error panel with a link back to / ; no separate error route | Exception flows |
| 5  | Responsive                       | Yes — PC (≥ 640px) and SP (< 640px, Tailwind sm breakpoint)                                        | §1              |
| 6  | Authentication required          | Yes                                                                                                | REQ-012         |
| 7  | Authorization / role restriction | Admin , Operator (DEC-001)                                                                         | REQ-012         |
| 8  | Applicable channel(s)            | Single web app — not applicable                                                                    |                 |

## 0-2. Page metadata (head)

The app's common head ( src/frontend/index.html : charset, viewport, favicon, bundled module script) is not restated.

### 0-2-1. Title

| No | Title                                                                                     |
|----|-------------------------------------------------------------------------------------------|
| 1  | Create mode: 新規製造指示 – ProductionManagementAI (New production order)                       |
| 2  | Edit mode: 製造指示 {order number} – ProductionManagementAI (Production order {order number}) |

### 0-2-2. Base

Not applicable — no <base> override.

### 0-2-3. Link

Not applicable — no screen-specific <link> tags; styles are bundled.

### 0-2-4. Meta

Not applicable — no screen-specific meta tags (internal, authenticated screen; no SEO metadata needed).

### 0-2-5. Style

Not applicable — no inline <style> block; all styling via Tailwind classes.

### 0-2-6. Script

Not applicable — no extra <script> tags; app JS is bundled.

0-3. URL parameters

| No | Parameter name    | Content                                                | Required / Optional                          | Reference                                     |
|----|-------------------|--------------------------------------------------------|----------------------------------------------|-----------------------------------------------|
| 1  | id (path segment) | Production order identifier; present only in edit mode | required in edit mode; absent in create mode | Unknown/invalid id → not-found exception flow |

1. Layout and mockup

Layout-level sketch only; the rendered per-state mockup belongs in 001\_DD.

3 ホーム &gt; 製造指示一覧 &gt; 新規

## 4 新規製造指示

5 製造指示 PO-2026-00001 を登録しました。

6 指示番号

保存時に採番

7 ステータス

下書き

8 製品 \*

製品を選択

9 数量 \*

10 納期 \*

yyyy/mm/dd

11 備考

任意

12 0/500

13 登録日時 2026/09/18 10:12

14 最終更新 2026/09/20 09:30

13-14: edit mode only

15

キャンセル

16

保存

Item 7 is a select in edit mode; 8 and 9 are read-only unless the status is Draft. Dialog 17-19 (discard changes) is not drawn.

| Item No. | Region / element                    | Notes (behavior, condition)                                                                                                                                                                                                                                                                                                                                                        |
|----------|-------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        | App header with navbar              | Shared application header, specified once in 003_BD "Shared application header" (H-1-H-5, E-29-E-31, WI-004 DEC-016): app name, navbar (「ダッシュボード」 (Dashboard), 「製造指示一覧」 (Production orders), 「新規製造指示」 (New production order); current entry marked), user and 「ログアウト」 (Sign out), a 「メニュー」 (Menu) button on SP. Current entry here: 「新規製造指示」 on create mode; 「製造指示一覧」 on edit mode |
| 2        | Signed-in user + 「ログアウト」 (Sign out) | Shared header (003_BD H-3; WI-001 behavior unchanged)                                                                                                                                                                                                                                                                                                                              |
| 3        | Breadcrumb                          | Last segment 「新規」 (New) (create) or the order number (edit)                                                                                                                                                                                                                                                                                                                        |
| 4        | Page heading                        | Text by mode, same as 0-2-1 minus the app suffix                                                                                                                                                                                                                                                                                                                                   |
| 5        | Message banner                      | Hidden unless there is a success or form-level error message; announced to assistive tech                                                                                                                                                                                                                                                                                          |
| 6–14     | Form fields                         | See §3; form card max width ~ 720px, centered                                                                                                                                                                                                                                                                                                                                      |
| 15       | 「キャンセル」 (Cancel) button             | Secondary style                                                                                                                                                                                                                                                                                                                                                                    |
| 16       | 「保存」 (Save) button                  | Primary style; disabled while saving; reads 「保存中...」 (Saving...) meanwhile                                                                                                                                                                                                                                                                                                         |
| 17–19    | Discard-changes dialog              | Modal, centered; shown only by E-07 when the form has changes (not drawn above)                                                                                                                                                                                                                                                                                                    |

## SP / mobile

1 PMAI

2 メニュー

3 < 製造指示一覧

4 新規製造指示

5 message banner

6 指示番号

保存時に採番

7 ステータス

下書き

8 製品 \*

製品を選択 ▼

9 数量 \*

10 納期 \*

yyyy/mm/dd

11 備考

任意

12 0/500

13 14 登録日時 / 最終更新 (edit mode)

16 保存

15 キャンセル

Same item numbers as PC. Differences: single column; quantity and due date stacked; buttons full width, 「保存」 above 「キャンセル」 ; breadcrumb collapses to a

back link; the navbar collapses behind the header's Menu button (003\_BD H-4, H-5).

## 2. Content block definition (CMS)

None — no externally managed content blocks.

## 3. Screen item definition

| No | Item (label)          | Variable name            | Control type                   | I/O | Data type  | Width / length | Initial value                                          | Placeholder                | Display condition                                                                       | Data source                     | Reference                           |
|----|-----------------------|--------------------------|--------------------------------|-----|------------|----------------|--------------------------------------------------------|----------------------------|-----------------------------------------------------------------------------------------|---------------------------------|-------------------------------------|
| 4  | Page heading          | —                        | label                          | O   | text       | —              | By mode (0-2-1)                                        | —                          | Always                                                                                  | Mode + <code>orderNumber</code> |                                     |
| 5  | Message banner        | —                        | label                          | O   | text       | —              | Hidden                                                 | —                          | After save success or form-level error                                                  | Save/load result                | §6 E-05, E-06                       |
| 6  | 「指示番号」 (Order number) | <code>orderNumber</code> | label                          | O   | text       | —              | Create: 「保存時に採番」 (Assigned on save); edit: saved value | —                          | Always                                                                                  | ProductionOrder.order number    | <code>PO-YYYY-NNNN</code> (DEC-012) |
| 7  | 「ステータス」 (Status)      | <code>status</code>      | select (edit) / label (create) | I/O | enum       | —              | Create: <code>Draft</code> (label); edit: saved value  | —                          | Always; select disabled when status is <code>Completed</code> or <code>Cancelled</code> | ProductionOrder.status          | §4 M-01, M-02; DEC-008              |
| 8  | 「製品」 (Product)        | <code>productId</code>   | select                         | I/O | identifier | —              | Create: none selected; edit: saved value               | 「製品を選択」 (Select a product) | Always; read-only unless status is <code>Draft</code> (REQ-018)                         | Product list (FN-004)           | §4 M-03                             |
| 9  | 「数量」 (Quantity)       | <code>quantity</code>    | textbox (numeric)              | I/O | integer    | up to 9 digits | Create: empty; edit: saved value                       | —                          | Always; read-only unless status is <code>Draft</code> (REQ-018)                         | ProductionOrder.quantity        |                                     |

| No | Item (label)           | Variable name | Control type | I/O | Data type | Width / length | Initial value                       | Placeholder        | Display condition                              | Data source                       | Reference                                                                                                                                    |
|----|------------------------|---------------|--------------|-----|-----------|----------------|-------------------------------------|--------------------|------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 10 | 「納期」<br>(Due date)     | dueDate       | date picker  | I/O | date      | —              | Create: empty;<br>edit: saved value | —                  | Always; editable in every status (DEC-004)     | ProductionOrder.due date          |                                                                                                                                              |
| 11 | 「備考」<br>(Notes)        | notes         | textarea     | I/O | text      | ≤ 500 chars    | Create: empty;<br>edit: saved value | 「任意」<br>(Optional) | Always; editable in every status (DEC-004)     | ProductionOrder.notes             |                                                                                                                                              |
| 12 | Notes counter          | —             | label        | O   | text      | —              | "0/500"                             | —                  | Always                                         | Live length of item 11            |                                                                                                                                              |
| 13 | 「登録日時」<br>(Created)    | createdAt     | label        | O   | datetime  | —              | Saved value                         | —                  | Edit mode only                                 | ProductionOrder.created timestamp | \$4 M-04                                                                                                                                     |
| 14 | 「最終更新」 (Last updated)  | updatedAt     | label        | O   | datetime  | —              | Saved value                         | —                  | Edit mode only                                 | ProductionOrder.updated timestamp | \$4 M-04                                                                                                                                     |
| 15 | 「キャンセル」<br>(Cancel)    | —             | button       | I   | —         | —              | —                                   | —                  | Always                                         | —                                 | \$6 E-07                                                                                                                                     |
| 16 | 「保存」<br>(Save)         | —             | button       | I   | —         | —              | —                                   | —                  | Always; disabled while a save is in progress   | —                                 | \$6 E-04                                                                                                                                     |
| 17 | Discard-changes dialog | —             | modal dialog | O   | text      | —              | Hidden                              | —                  | After Cancel when the form has unsaved changes | —                                 | Title 「変更を破棄しますか？」 (Discard your changes?), body 「この製造指示の変更は保存されていません。」 (Your changes to this production order haven't been saved.); REQ-019 |
| 18 | 「破棄」<br>(Discard)      | —             | button       | I   | —         | —              | —                                   | —                  | Inside dialog 17                               | —                                 | \$6 E-09                                                                                                                                     |

| No | Item (label)               | Variable name | Control type | I/O | Data type | Width / length | Initial value | Placeholder | Display condition                        | Data source | Reference |
|----|----------------------------|---------------|--------------|-----|-----------|----------------|---------------|-------------|------------------------------------------|-------------|-----------|
| 19 | 「編集を続ける」<br>(Keep editing) | —             | button       | I   | —         | —              | —             | —           | Inside dialog 17; receives initial focus | —           | \$6 E-09  |

Required fields (8, 9, 10) are marked with **\*** and **aria-required**, with the legend 「は必須項目です。」 (marks a required field). Once the order leaves **Draft**, items 8 and 9 carry the hint 「下書き以外の製造指示では変更できません。」 (Can't be changed once the order leaves Draft).

#### 4. Item value mapping

| No   | Item                | Source value                               | Displayed value                                                                                                              | Reference               |
|------|---------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| M-01 | (7) Status          | Draft / InProgress / Completed / Cancelled | 「下書き」 / 「進行中」 / 「完了」 / 「取消」 (Draft / In progress / Completed / Cancelled)                                                    | WI-005 DEC-004          |
| M-02 | (7) Status options  | Current status                             | Draft → {下書き, 進行中, 取消}; InProgress → {進行中, 完了, 取消}; Completed → {完了} (disabled); Cancelled → {取消} (disabled). Labels as M-01 | REQ-017, DEC-008        |
| M-03 | (8) Product         | Product ID                                 | "{SKU} — {name}", e.g. P-1004 – ドライブシャフト                                                                                     | DEC-005; WI-005 DEC-002 |
| M-04 | (13)(14) Timestamps | UTC timestamp                              | YYYY/MM/DD HH:mm, 24-hour, in the browser's timezone, after the label: 「登録日時 2026/09/18 10:02」                               | WI-005 DEC-006          |

#### 5. Validation rules

Client-side checks give early feedback; the server repeats every check and is authoritative. Error message texts are drafts; final IDs/wording are fixed in 001\_DD.

| No   | Item         | Check content                         | Validation rule              | Check condition                 | Error message (ID)                                                                                                                            | Reference        |
|------|--------------|---------------------------------------|------------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| V-01 | (8) Product  | Required; must be an existing product | Not empty; exists in Product | Create; edit while <b>Draft</b> | MSG-E001 「製品を選択してください。」 (Select a product.) / MSG-E002 「選択した製品は存在しません。」 (The selected product no longer exists.)                              | REQ-015          |
| V-02 | (9) Quantity | Required positive whole number        | Integer 1–999,999,999        | Create; edit while <b>Draft</b> | MSG-E003 「1以上の整数を入力してください。」 (Enter a whole number of 1 or more.) / MSG-E010 「数量は999,999,999以下で入力してください。」 (Quantity can't exceed 999,999,999.) | REQ-013, DEC-024 |

| No   | Item                     | Check content                            | Validation rule                                                                                                              | Check condition                                                                  | Error message (ID)                                                                                                                     | Reference                 |
|------|--------------------------|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| V-03 | (10) Due date            | Required valid date                      | Not empty; valid date                                                                                                        | Always                                                                           | MSG-E004 「納期を入力してください。」 (Enter a due date.)                                                                                            | REQ-014                   |
| V-04 | (10) Due date            | Today or later                           | ≥ today, where "today" is the date in the configured plant timezone, <code>Asia/Tokyo</code> (DEC-011, DEC-017)              | On create; on edit only when the due date differs from the saved value (DEC-009) | MSG-E005 「納期に過去の日付は指定できません。」 (Due date can't be in the past.)                                                                          | REQ-014, DEC-009, DEC-011 |
| V-05 | (11) Notes               | Length                                   | ≤ 500 characters                                                                                                             | Always                                                                           | MSG-E006 「備考は500文字以内で入力してください。」 (Notes can't exceed 500 characters.)                                                                   | REQ-016                   |
| V-06 | (7) Status               | Allowed transition                       | New status is the current status or a valid next state (M-02)                                                                | Edit                                                                             | MSG-E007 「このステータス変更は許可されていません。」 (This status change isn't allowed.)                                                                    | REQ-017                   |
| V-07 | (8)(9) Product, Quantity | Locked fields unchanged                  | If saved status ≠ <code>Draft</code> , product and quantity must equal saved values; otherwise the whole request is rejected | Edit                                                                             | MSG-E008 「下書き以外の製造指示では、製品と数量を変更できません。」 (Product and quantity can't be changed after the order leaves Draft.)                           | REQ-018, DEC-007          |
| V-08 | Whole order              | Not changed by someone else since loaded | Saved version equals the version the form was loaded with                                                                    | Edit (server only)                                                               | MSG-E009 「この製造指示は他のユーザーによって変更されました。再読み込みして最新の内容を確認してください。」 (This order was changed by someone else. Reload to see the latest version.) | REQ-011, DEC-010          |

## 6. Item events

| No   | Item                  | Event    | Event content                                                                                                                                                                                               | Reference        |
|------|-----------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| E-01 | Screen                | load     | Create: load products, show empty form. Edit: load order and products; on unknown <code>id</code> show not-found panel; on 401 go to <code>/login</code> ; show a loading indicator until both loads finish | FN-003, FN-004   |
| E-02 | (8)(9)(10)(11) fields | blur     | Run that field's client-side checks (§5) and show/clear its inline error                                                                                                                                    | FN-005           |
| E-03 | (11) Notes            | input    | Update counter (12)                                                                                                                                                                                         |                  |
| E-04 | (16) Save             | click    | Run all client checks; if any fail, show inline errors and move focus to the first invalid field; otherwise send the request, disable Save, and handle the result (E-05/E-06)                               | FN-001, FN-002   |
| E-05 | (16) Save — success   | response | Create: navigate to <code>/production-orders/{new id}</code> and show success banner. Edit: refresh the form with saved values (including status options and lock state) and show success banner            | REQ-010, REQ-011 |
| E-06 | (16) Save — failure   | response | Field errors → inline under each field; form-level errors (transition, lock, not found, server error) → error banner (5); 401 → <code>/login</code> ; entered values are kept                               | Exception flows  |

| No    | Item                                | Event          | Event content                                                                                                                                                                                                                                 | Reference               |
|-------|-------------------------------------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| E-07  | (15) Cancel                         | click          | If any field differs from the loaded (edit) or initial (create) values, open dialog 17. Otherwise navigate to <code>/production-orders</code> (SCR-002) without saving                                                                        | REQ-019, DEC-018        |
| E-08  | (7) Status                          | change         | Only selectable options per M-02; no immediate save — takes effect on Save                                                                                                                                                                    | DEC-008                 |
| E-07a | Navbar, breadcrumb or app-name link | click / Enter  | If the form has unsaved changes (as E-07), cancel the navigation and open dialog 17, remembering the link's destination. Otherwise navigate normally. Browser Back, reload and closing the tab are not guarded (unchanged; WI-004 DEC-022)    | REQ-019; WI-004 DEC-022 |
| E-09  | (18) Discard / (19) Keep editing    | click / Escape | Discard: close the dialog and navigate — to the remembered destination when opened by E-07a, otherwise to <code>/production-orders</code> — without saving. Keep editing or Escape: close the dialog, return focus to Cancel, keep all values | REQ-019                 |

## 7. External identity linkage

None — no external identity linkage.

## Actions and business rules

| Action                  | Trigger                                   | Business rule                                                                                                                                                                                                                                                                                   | Related requirement ID           |
|-------------------------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| Open screen / call API  | Any request to SCR-001 or its API         | Requires a valid session (else 401 → <code>/login</code> ) and role <code>Admin</code> or <code>Operator</code> (else 403) — enforced server-side, not only in the UI                                                                                                                           | REQ-012                          |
| Create order            | Save in create mode                       | All V-01–V-05 pass; order gets a system-generated unique order number <code>P0-YYYY-NNNNN</code> (year of creation in the <code>Asia/Tokyo</code> plant timezone, sequence restarts yearly — DEC-012) and status <code>Draft</code> ; created/updated timestamps set                            | REQ-010, REQ-013–REQ-016         |
| Edit order              | Save in edit mode                         | Applicable V-01–V-08 pass; only editable fields for the saved status are changed; updated timestamp set                                                                                                                                                                                         | REQ-011, REQ-013–REQ-018         |
| Change status           | Save with a different status              | Only <code>Draft→InProgress</code> , <code>Draft→Cancelled</code> , <code>InProgress→Completed</code> , <code>InProgress→Cancelled</code>                                                                                                                                                       | REQ-017                          |
| Record completion       | Save of <code>InProgress→Completed</code> | The order's completion time is set to the time of this save, in the same save as the status change; server-set, never entered or shown on SCR-001, never changed afterwards ( <code>Completed</code> is terminal). A rejected save records nothing. Consumed by the dashboard, SCR-003 (003_BD) | WI-004 REQ-033                   |
| Change product/quantity | Save in edit mode                         | Allowed only while saved status is <code>Draft</code> ; otherwise the whole request is rejected                                                                                                                                                                                                 | REQ-018, DEC-007                 |
| Cancel / leave          | Cancel button, or any in-app link         | Unsaved changes are discarded only after the user confirms                                                                                                                                                                                                                                      | REQ-019, DEC-018; WI-004 DEC-022 |

## Success and exception flows

| Flow                             | Trigger condition                                                 | System behavior                                         | Resulting state                                                                           |
|----------------------------------|-------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Success — create                 | Valid input, authorized user                                      | Order saved as <code>Draft</code> with new order number | SCR-001 edit mode for the new order, success banner                                       |
| Success — edit                   | Valid input and transition                                        | Changes saved                                           | SCR-001 edit mode with saved values, success banner                                       |
| Exception — unauthenticated      | No valid session (including expiry mid-edit)                      | 401, nothing saved                                      | Redirect to <code>/login</code>                                                           |
| Exception — forbidden            | Signed in without <code>Admin / Operator</code> role              | 403, nothing saved                                      | Error panel 「製造指示を管理する権限がありません。」 (You don't have permission to manage production orders.) |
| Exception — validation           | Any V-01–V-05 fails                                               | Rejected, nothing saved                                 | Inline field errors; values kept                                                          |
| Exception — invalid transition   | V-06 fails (e.g. crafted request)                                 | Rejected, nothing saved                                 | Error banner; status unchanged                                                            |
| Exception — locked field changed | V-07 fails                                                        | Whole request rejected (DEC-007)                        | Error banner; values kept                                                                 |
| Exception — order not found      | Edit route/API with unknown <code>id</code>                       | 404                                                     | Not-found panel with a link back to the list                                              |
| Exception — products unavailable | Product list empty or fails to load                               | Save not possible in create mode                        | Product select shows 「利用可能な製品がありません。」 (No products available.) / load error banner        |
| Exception — concurrent change    | V-08 fails: order was changed by someone else after it was loaded | Whole request rejected, nothing saved (DEC-010)         | Error banner MSG-E009 with a Reload action; entered values kept until reload              |
| Exception — server/network error | Unexpected failure                                                | Nothing assumed saved                                   | Error banner 「問題が発生しました。もう一度お試しください。」 (Something went wrong. Try again.); values kept     |

## Data design overview

- **ProductionOrder** — order number (system-generated, unique, `PO-YYYY-NNNNN`), reference to exactly one Product, quantity, due date, status (`Draft / InProgress / Completed / Cancelled`), notes, created/updated timestamps, and the completion time — present exactly when the status is `Completed` (added by WI-004, 003\_DB).
- **Product** — name, SKU; a minimal reference table seeded with 30 sample automobile parts with Japanese names, such as `P-1001` ブレーキキャリパー (brake caliper), no catalog UI (DEC-005, DEC-019; WI-005 DEC-002).

Relationship: many ProductionOrders → one Product (required). Column types, keys, constraints and order-number generation belong to the database-design document.

## External interfaces

None — only this application's own backend API.

## Non-functional requirements

- **Security:** the role check ( Admin / Operator ) gates state-changing actions and must be enforced server-side. No field is PII or a secret. CSRF: no extra anti-forgery token; SameSite=Lax cookie, no CORS policy and JSON-only request bodies (DEC-020).
- **Accessibility (WCAG 2.2 AA, ai/rules/frontend.md ):** every control has a visible label; required fields use aria-required ; inline errors are linked via aria-describedby ; the message banner uses a live region; focus moves to the first invalid field on failed save; the discard-changes dialog traps focus, starts on 編集を続ける (Keep editing) and closes on Escape; read-only (locked) fields remain focusable and announced as read-only; visible focus indicators and sufficient contrast; fully keyboard-operable (including the date picker).
- **Observability:** create/edit endpoints are traced and logged per project defaults; exact spans/metrics are specified in 001\_DD.
- **Language:** the UI is Japanese only ( <html lang="ja"> , Japanese font stack); every text comes from the frontend catalog (WI-005 DEC-001, DEC-007).
- Performance/availability: inherits project defaults.

## Open questions and linked DD

| Question                                                                                       | Linked DD section                     | Status                                                                                           |
|------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------|
| Does the due-date ≥ today rule apply when an existing order's past due date is left unchanged? | 001_DD validation                     | answered in decisions.md (DEC-009)                                                               |
| Concurrent edits: reject stale save or last write wins?                                        | 001_DD processing flow / API contract | answered in decisions.md (DEC-010); mechanism decided in 001_DB (DEC-014)                        |
| Which timezone defines "today"?                                                                | 001_DD validation / configuration     | answered in decisions.md (DEC-011)                                                               |
| Order-number format                                                                            | Database design (001_DB)              | answered in decisions.md (DEC-012); generation and overflow handling decided in 001_DB (DEC-013) |
| CSRF protection for the new endpoints                                                          | 001_DD security                       | answered in decisions.md (DEC-020)                                                               |

| Question                                                         | Linked DD section                             | Status                                     |
|------------------------------------------------------------------|-----------------------------------------------|--------------------------------------------|
| Plant timezone value; Cancel with unsaved changes; demo products | 001_DD configuration / events;<br>001_DB seed | answered in decisions.md (DEC-017–DEC-019) |